Project Title

CCSW 223 Lab Project Report

[your Section]

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Lamees Algamdi	2310331	Functional and non-Functional Requirements
Lama Kayyal	2380245	Use case
Layan Mohammed	2310983	schedule estimation
Ghadi Algamdi	2310345	The introduction

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1 Introduction

The technology provides real-time order tracking and different payment alternatives. It also allows customers to leave evaluations and ratings to assist improve services, as well as customize orders depending on their preferences. The system is accessible via online and mobile platforms, ensuring a quick and smooth buying experience.

2 Problem Definition

Some people face a problem in communicating their desires to cake shop owners. Accordingly, we designed an application through which the customer can design the cake he wants and makes it easy for the project owner to understand the customer's requirements easily.

3 Information Gathering

3.1 Interview Description

In this interview we are going to interview some potential users and customers about our Cake application, what do they think about the idea, is it really going to help them and what are the problems that they want to avoid while using it. We are also going to interview some real cake shop owners to discover their problems when dealing with customers, what do they do, and how do they think a cakes application is going to help them in managing the shop. And this is the questions that we prepared:

For the owner:

◦ What do you use to communicate with the costumer?

◦ What are the challenges or difficulties that faced you as a cake shop owner?

◦ How does this problem impact you?

◦ What features do you think are most important for cake order applications?

◦ Can you tell me how a cake application could help you manage special orders or custom cake requests more efficiently?

For the costumers:

◦ What's the problem that face you when you order a cake?

◦ If you have an allergy, did you find a cake shop that accepts your special request?

◦ How do you think a cake application is going to help you?

◦ What are you looking for in a cake shop application?

◦ What are the problems that you wanted to avoid, or you don’t want to find it in a cake application shop?

3.2 Interview Analysis

Interviewer	**Interviewee**	***Analyst Comment***
Yara	Nada Saeed (potential user) Doesn't have a job The date of the interview is 7 September	From all the information that I have gathered I discovered that most of the people that always order a cake has a similar problem which include the misunderstanding that could happen between the owner and the customer or not completing the customer needs because of lack of the

		necessary communication and listening from the owner side to the customer. Another thing that most of the customer wants is for the shop to accept their special requests like, print a specific photo in the cake and shows most of the ingredients for the people who maybe have allergies.
Yara	Second interviewee: Hala Khalid (potential user) A student The date of the interview is 10 September	My interviewee said that a cake's applications is going to help her for a lot of reasons, first, she can customize her cake the way she like it for any occasions she want, second, it would be easier for her to avoid any mistakes in the order since she can write all her needs in the description , and lastly it would be faster because you can choose the day and the time you want the cake to be in your hands and since there is many ways to pay like cash or cards or others. "Every customer wants an easy application to use" that's what she said when I asked her about the ideal application and she is right every customer want an application that is organized and easy to read and use, and every customer wants to avoid a non-working or unorganized application.
Lamees	First Interviewee: Aryam Alqarni (potential user) Job: Student The date of interview:9 September	Every time she orders a cake, she always seems to face the same issue. She never knows the production or expiration dates, and the cake's ingredients are often unclear. It becomes even more stressful when certain cakes contain ingredients that could trigger allergies for others. She believes a cake app would solve all these problems. It would allow her to choose exactly what she wants, including a full list of ingredients and flavors she likes, and even give her the option to select the perfect size. Whenever she browses the Cake factory app, she looks for quality, the right price, and a clear presentation of the cake. Plus, she prefers to avoid unnecessary external ingredients that aren't always wanted. The main issues she's trying to avoid are cakes without clear dates, missing ingredient lists, and the lack of price transparency.
Lamees	Second Interviewee: Bushra Alghamdi (potential user) Job: doesn't have The date of interview: 8 September	Whenever she orders a cake, she struggles with describing the design she wants. Although she doesn't have any allergies, she's noticed that some stores accept custom cake orders, while others don't. She believes a cake app would save her time and also allow her to accurately specify potential allergens for others. When using a Cake Factory app, she's searching for a design that matches the exact measurements she needs. One of the key issues she wants to avoid is the lack of precision in the cake's dimensions and any bias in design options when making custom orders.
Ghadi	First Interviewee: Amal Abdulrahman (potential user) Job: businesswomen The date of interview: 8 September	Ordering a cake often presents challenges, such as receiving a product that doesn't match expectations and managing allergy requests. An ideal cake app would simplify the process with an intuitive design, straightforward customization options, and clear communication for dietary needs. It should also provide a transparent experience, free from hidden fees and confusing features.

Ghadi	Second Interviewee: Sara Ali (potential user) Job: student The date of interview: 8 September	Issues with cake ordering include inconsistent quality and challenges with dietary needs. A well-designed cake app could resolve these by providing detailed customization, clear allergen information, and real-time updates. It should be transparent, easy to use, and reliable in addressing special requests to ensure a smooth and stress-free experience.
Lama	First Interviewee: Nouf Alharbi Job: businesswomen The date of interview: 6 September	From what I gathered from the interview with her, I found that she uses social media platforms to communicate with customers and faced a problem in delivering the order properly to the customer. She noted that the problem hindered her work and caused long interruptions, as well as promoting her store. She believes that the features she wants to have been that the promotion is for the target group and is not done randomly, as well as the ease of delivering the product safely. Regarding managing orders, she said that having an application helps her in making appointments with a specific delivery time, as well as notifying the customer when the quantity runs out and providing a delivery representative during working hours on an ongoing basis. According to what she said, she preferred to work online because it is less expensive and provides products in quantities according to demand.
Lama	Second Interviewee: Amani Saleh Job: businesswoman The date of interview: 8 September	Through the interview with her, I found that she uses social media platforms to communicate with the customer. She said that the problem she faced was in providing some of the customers' requirements in the form of the cake and its decoration. The problem did not affect her completely, but it led to the cancellation of orders. She added that the important features are the quality of the application and the provision of delivery. She noted that it will help her organize reservations during events and provide payment through it. As for her, she prefers to work online because it is more cost-effective, and the actual store requires a higher cost.
Toleen	First Interviewee: Haneen Tunkar (Potential user, business owner) Job: businesswomen The date of interview: 8 September	There are many difficulties and problems with customers for example, many customers ignore important side notes, such as putting the cake in the refrigerator. This ruins the shape of the cake, which causes the customer to get angry and write bad feedback, which tarnishes the owner's reputation in the job market. Also, the owners prefer to use a comprehensive application for all types of cakes clearly and in detail to facilitate the communication process between the two parties and reduce material and moral losses as much as possible. Can also set a specific limit for requests received by the application in one day and put information about the working hours during which the customer can communicate.
Toleen	Second Interviewee: Razan Tunkar (Potential user, business owner) Job: businesswoman The date of interview: 9 September	Most customers do not have flexibility when requesting a specific design or color for the cake, and never accept anything approximate to the desired shape. So, that it is important to put some features in the application that make it easier for him to understand the customer's special requests, such as specifying the type of design according to the type of party, whether he wants to write on the cake or not, putting a note on the ingredients that cause allergies, and other information that helps to understand the customer in an excellent way.
Layan	First Interviewee: Sara Hazem	She seems to have one major issue, and that is the lack of transparency in pricing, she frequently discovers that the ultimate

	(Potential user) Job: Student The date of interview: September 8th	expense is much higher than what she was expecting. Her allergies make it difficult to locate a store that genuinely understands her needs, even though some are flexible. Many bakeries fail to make it apparent if they may include specific requests. The ordering process might be made much easier with a cake app. It would save a ton of time and concern if it had features like allergen filtering choices, affordable prices, and direct communication with bakers.
Layan	First Interviewee: Nawaf Mohammed (Potential user) Job: unemployed The date of interview: September 9th	Based on all the information I received He claimed that often this process does not go as smoothly as he would want. Some cake apps, for instance, take a long time to confirm the order and deliver the cake, and even then, the cake doesn't look or taste like what was expected. He has an allergy, but stores don't give him enough clear information about it or won't let him ask for specific allergens like gluten or nuts to be removed. If there were more obvious choices for it, that would be great. The entire ordering process might be more Streamlined with a cake app. Having the ability to track the order, get in touch with the store, and make sure the consumer is satisfied would be extremely helpful

Through the interviews, we discovered the problems they faced in two aspects. The first is related to the ordering method, which serves the customer and the seller by specifying the time, day, and quick communication. The second is related to the quality of the cake provided in terms of size, shape, flavor, ingredients, and personal additions, in addition to the production and expiration dates.

4 Planning Phase

4.1 Project Goals

Fulfilling customer requirements accurately without errors. Thus, gaining more application users and increasing profit by 5% in less than six months

4.2 Cost and schedule estimation

The budget will reach approximately 150,000 SA

Project Estimated Schedule:

Tasks	Estimated duration	Start date	End date
Meet with stakeholders to gather requirements (e.g., customization features, delivery options, payment methods).	4 days	28/8/2024	31/8/2024
Assess the technical and financial feasibility of the project.	1 day	1/9/2024	2/9/2024
Create the project plan and set key milestones.	2 days	2/9/2024	4/9/2024
Design user interface and user experience wireframes for the app	1 week	7/9/2024	7/9/2024
Review the designs with stakeholders and make necessary adjustments.	3 days	15/9/2024	18/9/2024
Design the database structure for orders, users, cake options.	2 days	20/9/2024	20/9/2024
Set up the frontend framework	2 days	23/9/2024	25/9/2024
Develop the homepage and the cake customization interface.	1 week	25/9/2024	2/10/2024
Implement user registration, login, and profile features.	1 week	2/10/2024	9/10/2024
Implement order history and integrate payment gateways.	1 week	9/10/2024	16/10/2024
Set up the server and configure the database.	1 week	18/10/2024	25/10/2024
Develop APIs for cake customization, orders, and delivery tracking.	2 weeks	25/10/2024	8/11/2024
Integrate payment gateway and set up delivery scheduling features.	1 week	8/11/2024	15/11/2024
Implement user management features and set up email/SMS notifications.	1 week	15/11/2024	22/11/2024
Perform unit tests on both frontend and backend components.	1 week	24/11/2024	1/12/2024
Conduct integration testing to ensure smooth interaction between frontend and backend.	1 week	1/12/2024	8/12/2024
Allow stakeholders and test users to evaluate the app's functionality.	1 week	8/12/2024	15/12/2024
Submit the app to the respective app stores.	3 days	20/12/2024	23/12/2024
Monitor app performance and fix any issues after launch.	1 week	23/12/2024	30/12/2024

4.3 The Feasibility Study

A- Problem Definition: The current systems struggle to efficiently handle the increasing demand for cake orders, leading to customer dissatisfaction

B- Scope Objectives of “new system”: Our proposed name for our application is Celestial Cakes. This app is designed to provide a quick and easy way to order a cake or customize it by the user himself then deliver it in the time and place the user chooses, saving the user time and hard work and that’s what makes our application special. Additionally, it offers high-quality service while efficiently managing increasing demand. By utilizing an effective system, the app will be safeguarded against the risks of lagging or getting hacked no

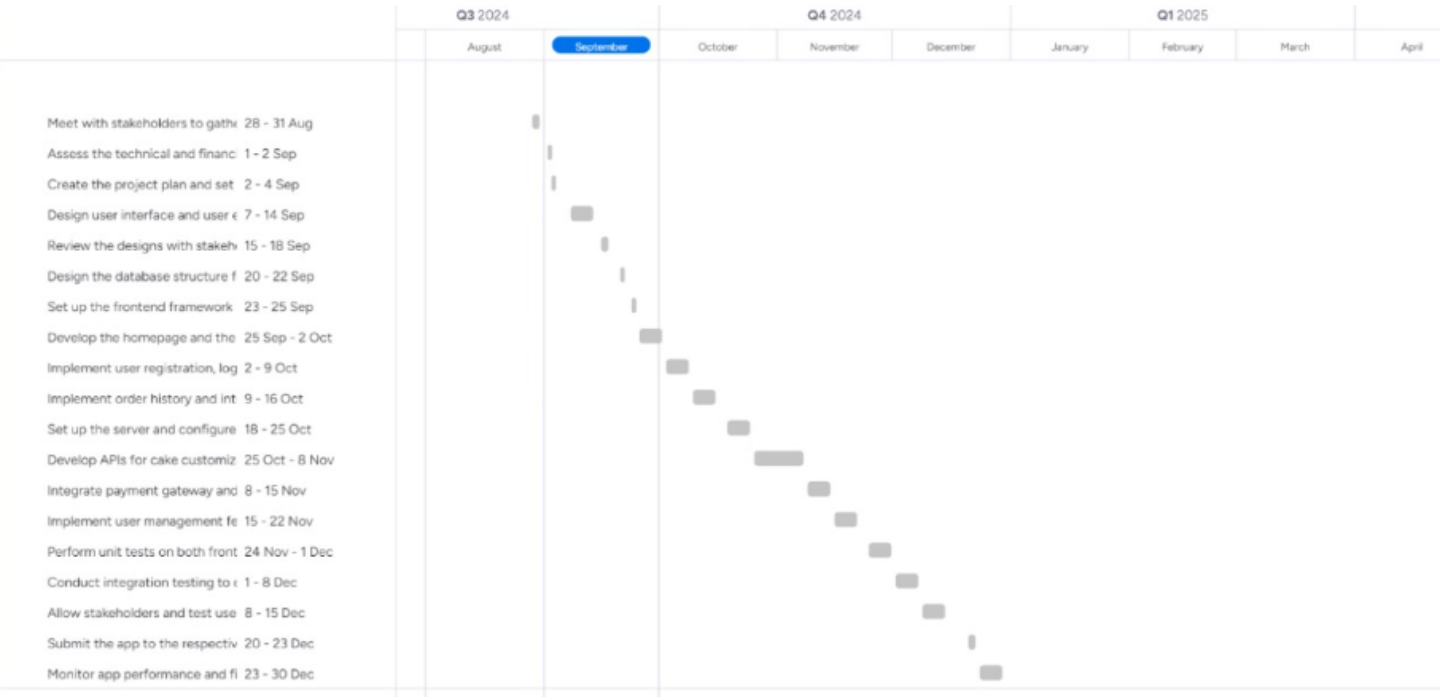
C- Alternative Solutions:

1- Create a new system.

2- Modify or add features to an existing system.

3- Do nothing.

4.4 Project Plan



5 Analysis Phase

5.1 Stakeholders

A- The Client

The cake shop owner how need for an application for the customer to customize their cakes and deliver it to them.

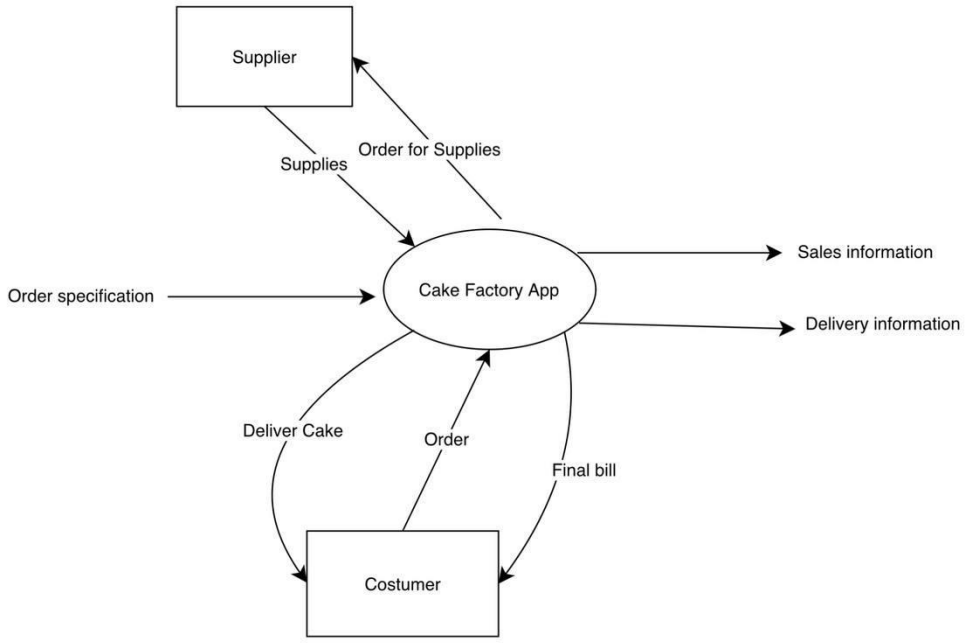
B- The Customer

The Individuals how are looking to buy a cake for any special occasions such a birthday, weddings parties and other occasions. Also, for companies that need cakes for meetings and special events. Or maybe restaurants and cafes that want to offer special cakes to their customers, and stores that want to sell cakes as part of their products.

(Examples: the book page 104).

1. End Users: Customers who use the app to order and customize cakes.
2. System Owners: who own the app and define its business goals.
3. Owner assistant: who receive and help the owner with the orders and necessary arrangements.
4. Bakers: who are responsible for preparing and decorating the cakes according to customer specifications.
5. System Managers: Individuals responsible for the app's operation and maintenance.
6. Payment processors: who handle the secure processing of customer payments and refunds.
7. Customer service: who manage customer requirement. complaints, and ensure smooth processing of orders.
8. External Stakeholders: Third parties affected by the app, such as suppliers and delivery services.

5.2 Context Diagram



5.3 Event Table

Event Name	Input and Output	Summary
1. Receiving order	Supplies (in)	Receiving ingredient from a supplier
2. Ordering a supplier	Order for supplies (out)	Order ingredient from a supplier
3. Payment	Payment (in)	The customer does the payment to the cake shop app
4. Customer Ordering	Order (in)	The customer orders cake from the cake shop app
5. Final bill Generation	Final bill (Out)	The customer received the final bill from the cake shop app
7. Delivering cake	Deliver cake (Out)	The customer received the cake
8. Receiving specification	Order specification (in)	The restaurant Received the specification from customer.
9. Delivery information Report	Delivery information(out)	The system produces a delivery report to a user

5.4 Functional Requirements

FR1: The system shall enable customers to browse and filter cake options by flavor, size, and occasion.

FR2: The system shall allow customers to customize cakes with decorations and special requests.

FR3: The system shall have the capability to allow customers to write special notes in a designated area for free-text input on the cake.

FR4: The system shall include a search feature for customers to quickly find specific cakes.

FR5: The system shall enable customers to share their cake selections on social media.

FR6: The system shall enable customers to see the final image of the cake in 3D before placing the order.

FR7: The application shall allow users to save cake designs for later review.

FR8: The system shall allow the user to specify the date and location of delivery.

FR9: The system shall send notifications to the user to track the status of the order.

FR10: The system shall notify the customer when the items are out of stock.

FR11: The system shall allow the owner to add new products.

FR12: The system shall allow the customer to add products to the cart.

FR13: The system notifies the owner of a new order.

FR14: The system shall allow customers and owner to log in and out in any time.

FR15: The system shall allow the customer to identify components that cause allergies or should be avoided.

FR16:The system shall allow the user to save their preferences and previous orders on their account.

FR17:The system shall allow both customer and owner texting.

FR18:The system shall allow the owner to modify information about their products.

5.5 Non-Functional Requirements

Performance

1. The application should respond to user interactions (customizing a cake or checking delivery options) within 2 seconds under normal load conditions.
2. The system should be able to handle at least 500 users at one time without degradation in performance.

Security

1. All user data, including personal information and payment details, must be encrypted both in transit and at rest to prevent unauthorized access.

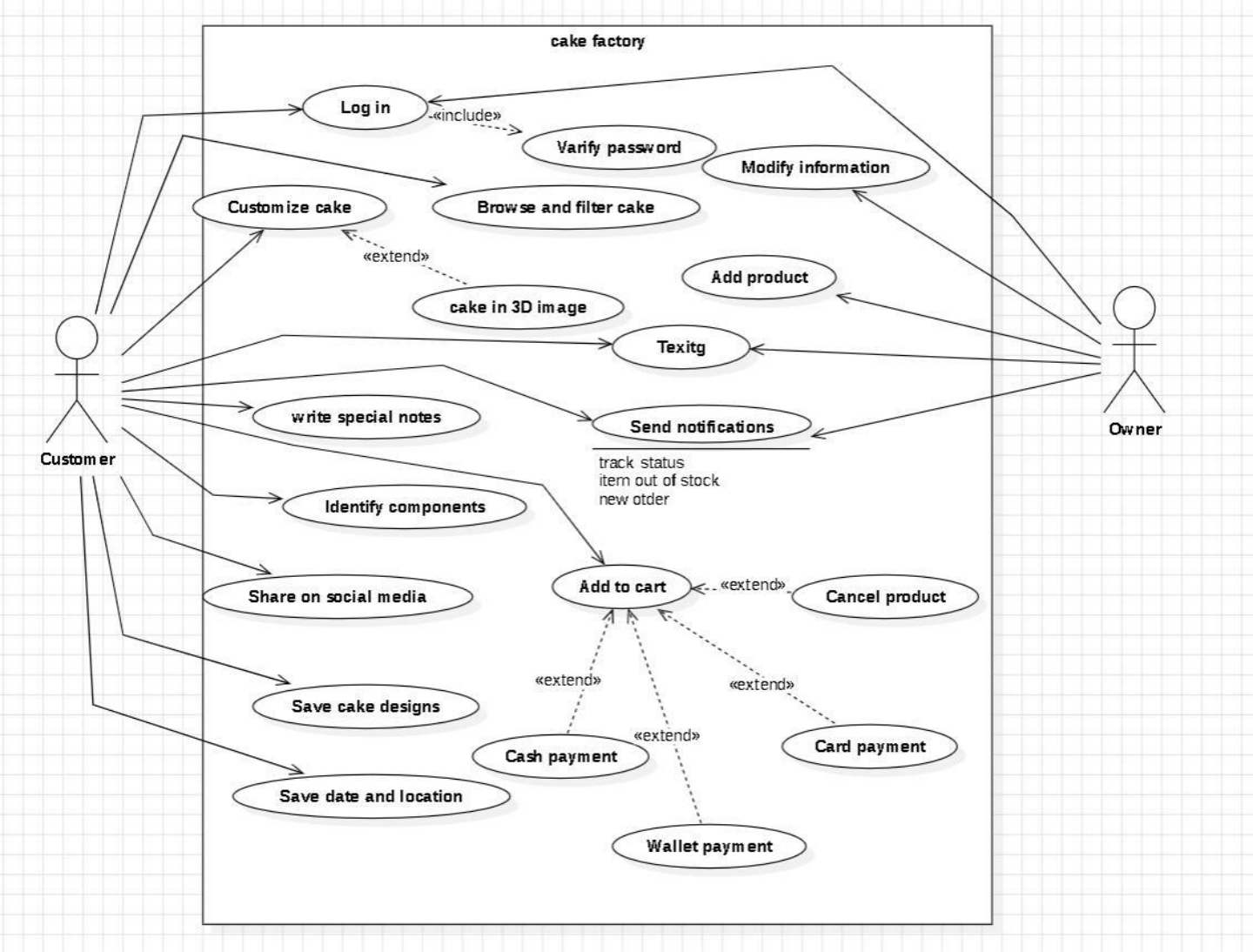
2. The application should implement multi-factor authentication for user accounts to enhance security during login.

Availability

1. The application must maintain 99.9% uptime, ensuring it is always accessible to users except during scheduled maintenance.

2. The system should have a disaster recovery plan in place to restore functionality within 4 hours of a critical failure.

5.6 Use case Diagram



5.7 Scenarios

Name	Log in
Id	UC1
Actors	Customers
Precondition	customers must have an account so you can be able to sign into it
Path	1. Create an account that you don't have and fill it up with all your information like the full name, the email, the address, the password and so on. 2. After you create an account sign in using the email and the password 3. A verification message with a code will appear in your email to make sure that you are the one who requested to sign in 4. After that you will be signed in and you can use the application freely

Name	**Specify Delivery Date and Location**
Id	UC2
Actors	Customers
Precondition	The customer must have an active order.
Path	1. The customer goes to the checkout page. 2. The system asks the customer to choose a delivery date. 3. The customer selects a suitable date from the calendar. 4. The system asks the customer to enter the delivery location or confirm a saved address. 5. The customer enters a new address or selects a saved one. 6. The system checks if the date and location are valid. 7. The system confirms the delivery details and moves to payment or order confirmation.

Name	**Order status notification**
Id	UC3
Actors	customers
Precondition	Customers must place an order to get the notification.
Path	1-login 2-place an order. 3- confirmed the order. 4-the system asks the user to agree on getting notifications for order status. 5- the system asks you to choose preferred notifications method such as (SMS/email/in app). 6- users click on their preferred method. 7- the user receives notifications to track the status of the order.

Name	**saving preferences and previous orders**
Id	UC4
Actors	customers
precondition	--

path	1-login 2-set preferences (from the profile section, select add favorite cake, add your preferences, save the details) 3-view saved preferences 4-reorder Cakes (click the order history within the profile) 5-Receive Suggestions- Check the “Recommendations” section for suggestions based on previous saved preferences.

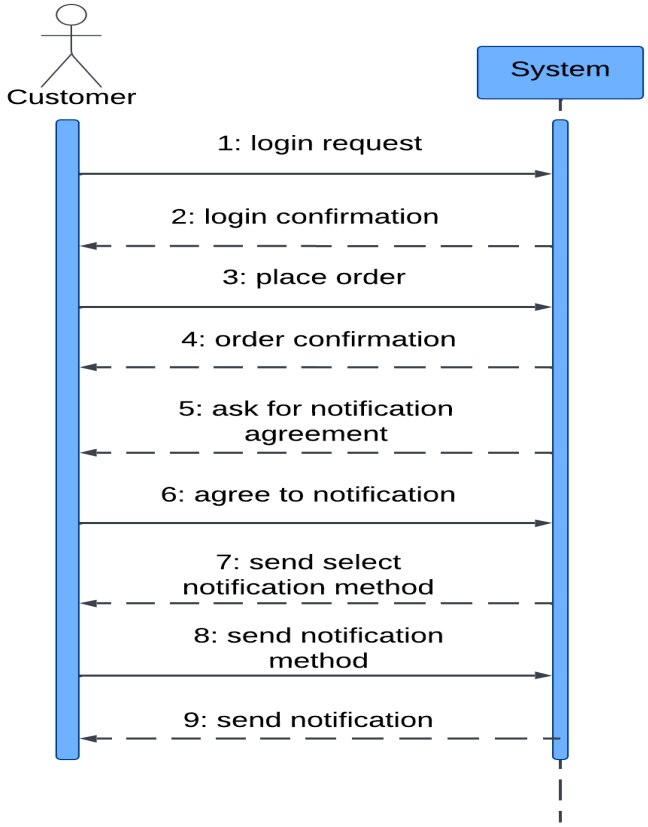
Name	**notify of new order**
Id	UC5
Actors	owner
Precondition	owner must log in Path
Path	1-the system notifies the owner about new order 2-owner press on the notification 3-owner check on the order 4-accept the order

Name	**Add To Cart**
Id	UC6
Actors	Customers
Precondition	The customer must log in the system
Path	1- customer select product by clicks on a product for more details. 2- The customer selects any necessary options (size, color, etc.). 3- The customer clicks the "Add to Cart" button. 4- The customer can view the cart to confirm items. 5- The customer decides to either add more items or proceed to checkout

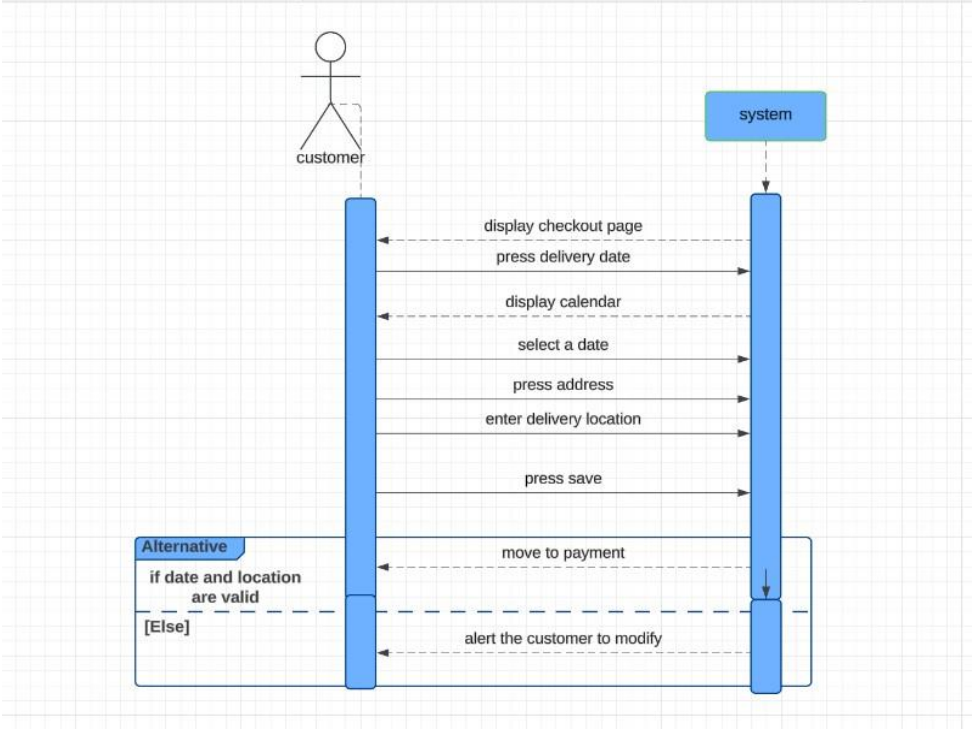
6 Design Phase

6.1 System Sequence Diagram

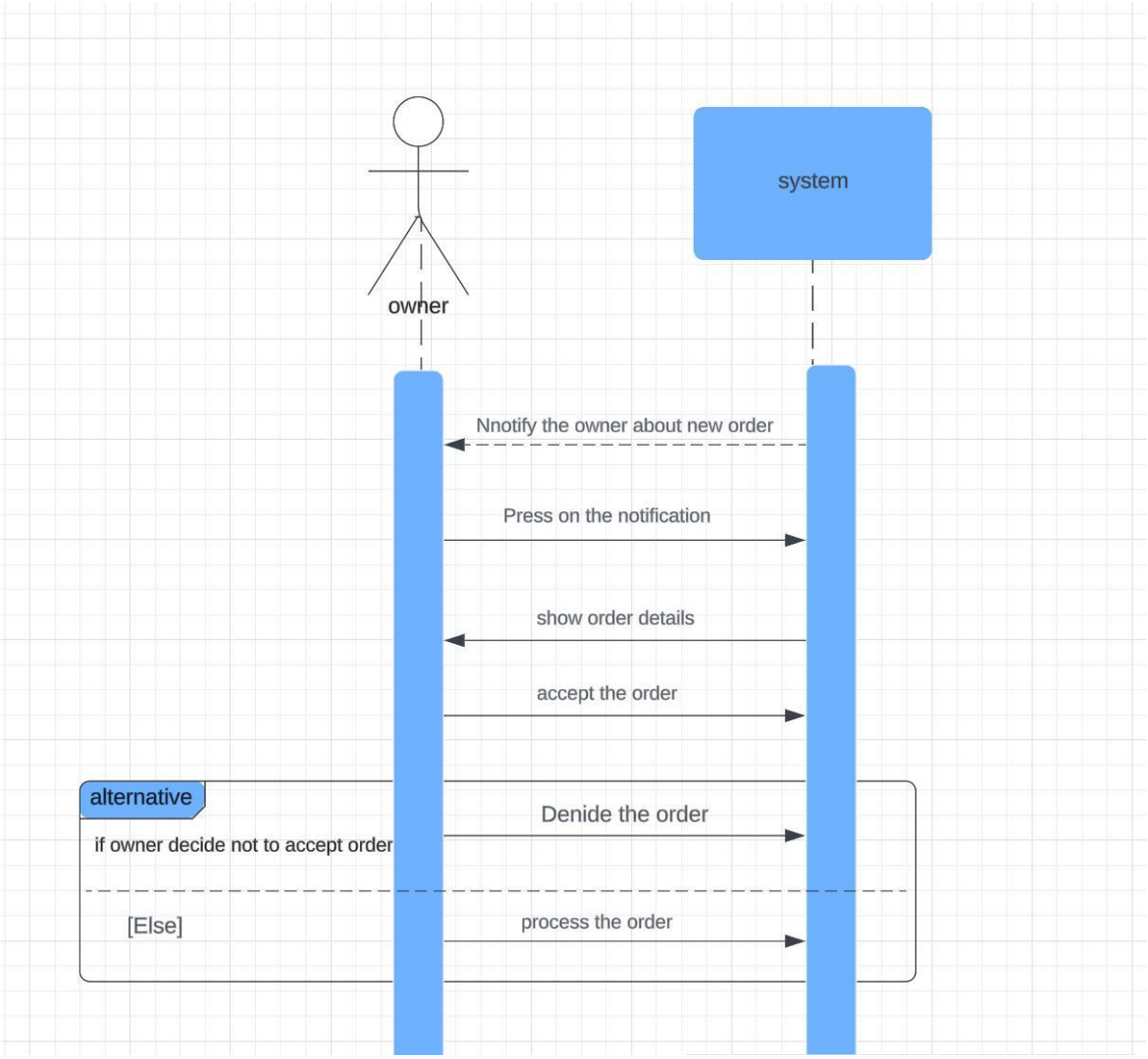
Sequence Diagram#1: For use case ”Order status”



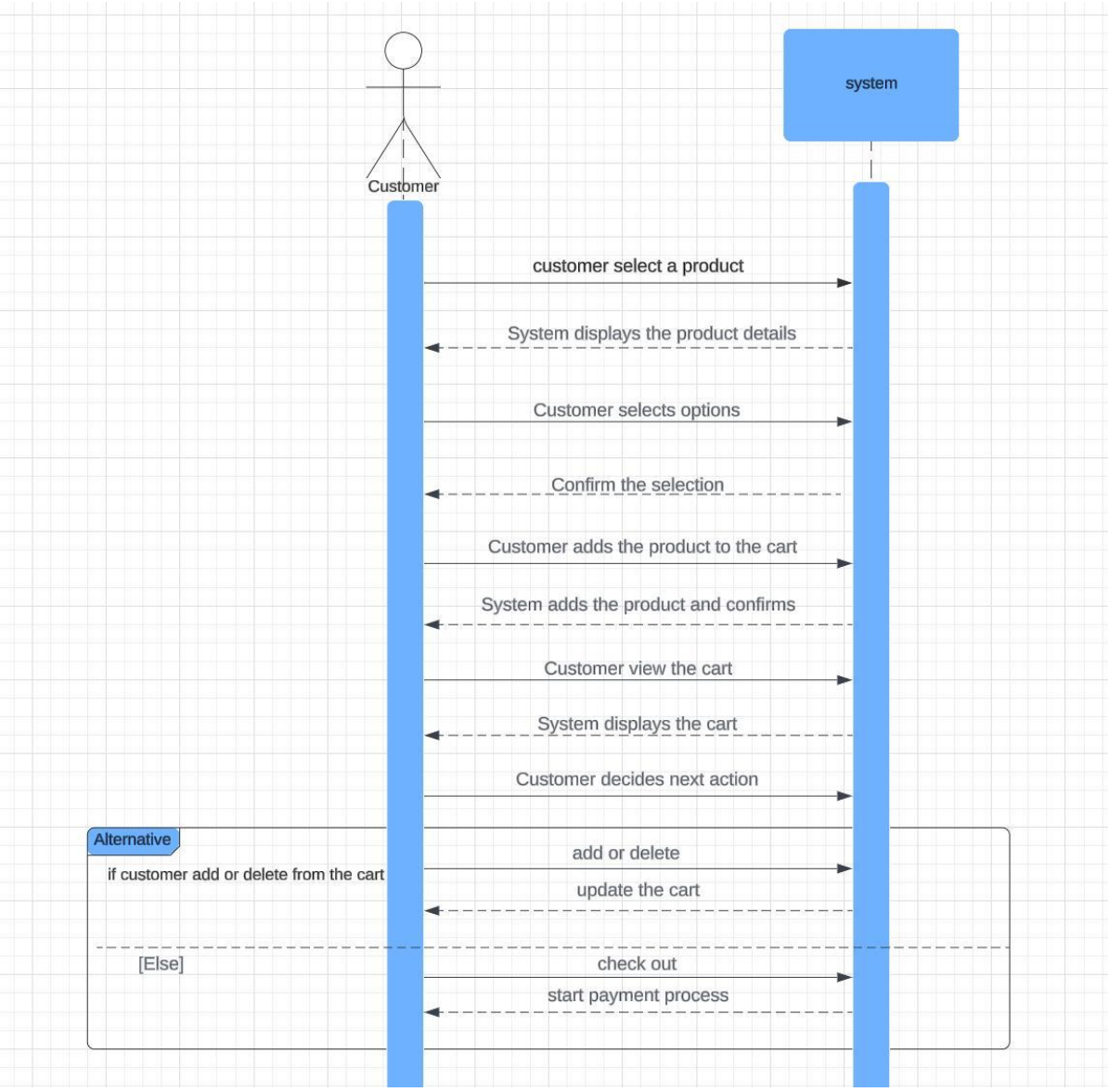
Sequence Diagram#2: For use case ” Delivery date and Location”



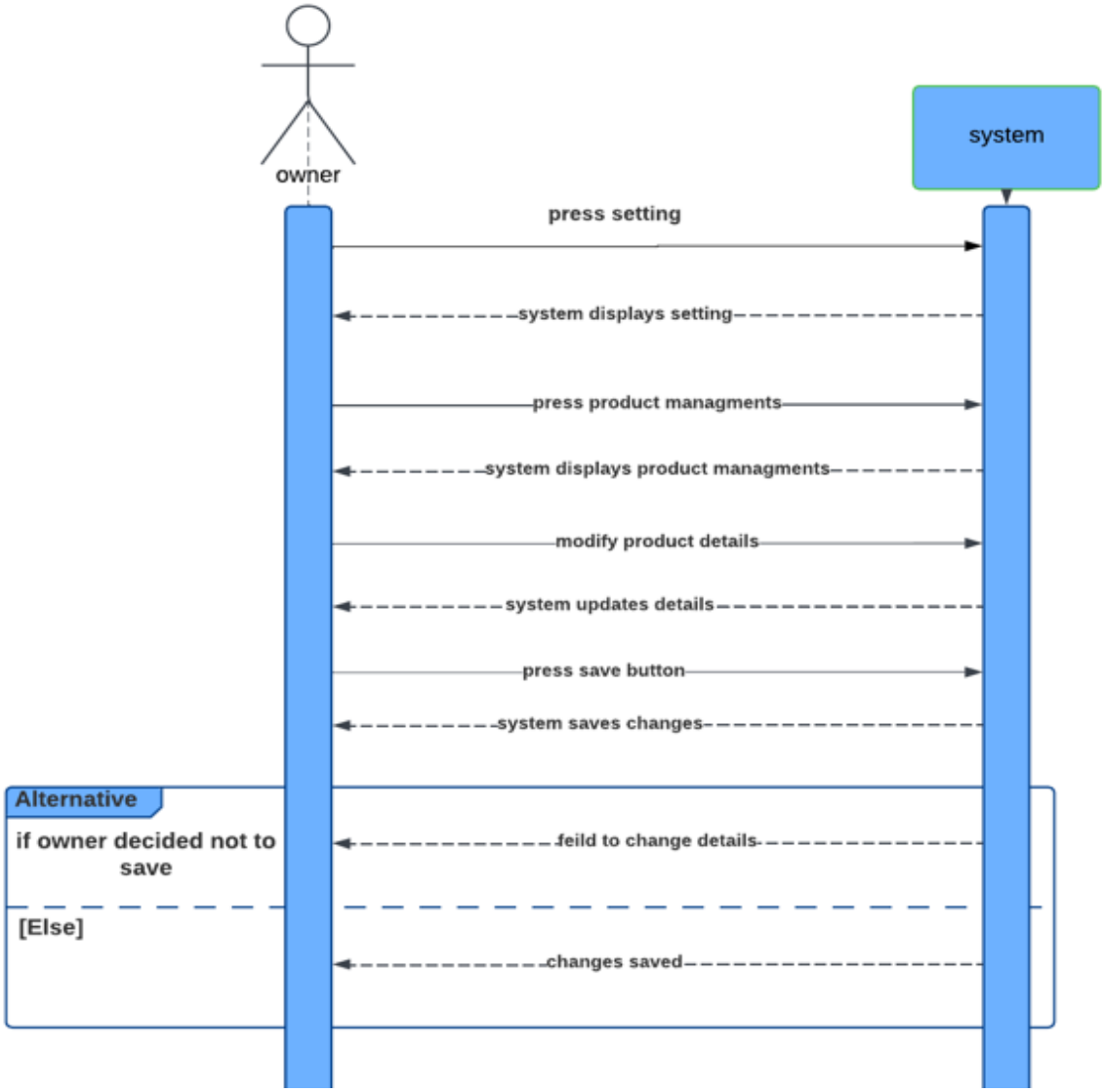
Sequence Diagram#3: For use case ” Notify new order”



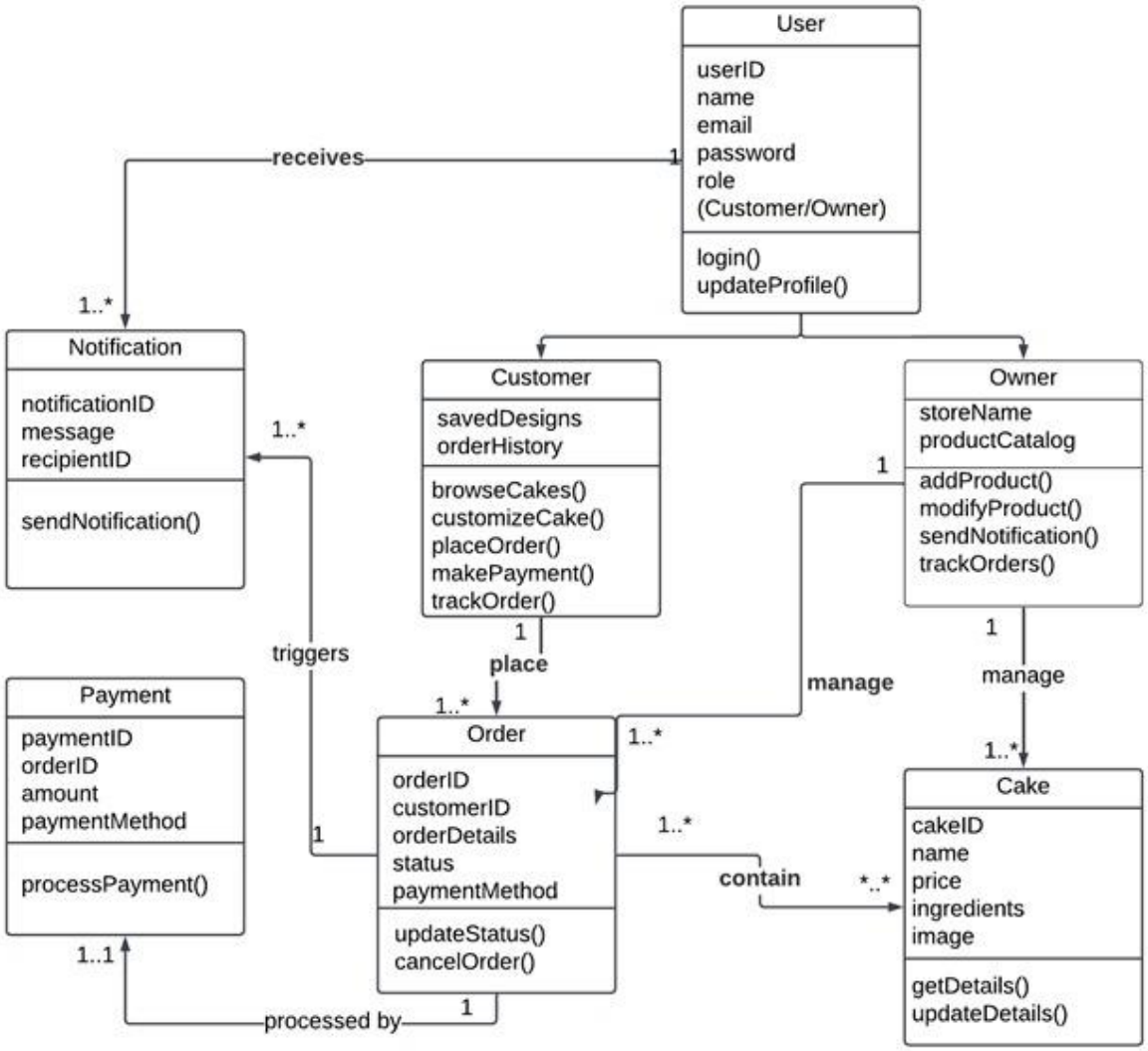
Sequence Diagram#4: For use case " Add to cart"



Sequence Diagram#5: For use case" Modify Information"



6.2 Class Diagram



7 Conclusion

To summarize, this system intends to improve the cake selling process by offering a technological solution that streamlines order administration, inventory monitoring, and sales analysis. Technology has effectively streamlined everyday operations, saving time and effort for both consumers and business owners. It also enhances order accuracy and decreases human mistakes, resulting in improved client experiences and more reliability.

With continued development, the system may be expanded to incorporate features like a customer loyalty program, payment method integration, and online purchase support. These improvements will improve the system's capacity to satisfy rising market demand and fulfill company objectives more effectively